MATHS MANTHAN
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 08** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 11: Areas Related to Circles** **MM: 30**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MULTIPLE CHOICE

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Q1. A circular disc of radius 10 cm is divided into 8 equal sectors. The central angle of each sector is: [1]

(A) $30°$ (B) $45°$ (C) $60°$ (D) $75°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Divide Total Central Angle [0.5 Mark]

A full circular rotation subtends $360°$ at the centre:

$$\theta = \frac{360°}{n} = \frac{360°}{8} = 45°$$

Correct Option: (B) $45°$

Central Angle $\theta = \mathbf{45°}$

Q2. A square of side 10 cm has an incircle of radius r and a circumcircle of radius R. The ratio of their areas is: [1]

(A) $1:2$ (B) $1:\sqrt{2}$ (C) $1:4$ (D) $2:3$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Radii of Incircle and Circumcircle [0.5 Mark]

For a square of side $a = 10$ cm:

$$\text{Inradius } r = \frac{a}{2} = 5 \text{ cm}, \quad \text{Circumradius } R = \frac{a\sqrt{2}}{2} = 5\sqrt{2} \text{ cm}$$

Step 2: Ratio of Their Areas [0.5 Mark]

$$\frac{\text{Area(Incircle)}}{\text{Area(Circumcircle)}} = \frac{\pi r^2}{\pi R^2} = \left(\frac{r}{R}\right)^2 = \left(\frac{5}{5\sqrt{2}}\right)^2 = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2} = 1:2$$

Correct Option: (A) $1:2$

Ratio of Areas $= \mathbf{1:2}$

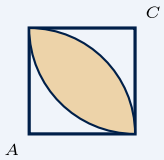
Q3. In a square $ABCD$ of side 7 cm, two quadrants are drawn with centres A and C and radius 7 cm. The area of the shaded leaf region between them is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 28 cm^2 (B) 14 cm^2 (C) 35 cm^2 (D) 21 cm^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Formula for Leaf Area [0.5 Mark]

The overlapping area of two opposite quadrants in a square:



$$\text{Area} = 2 \times \left(\frac{1}{4}\pi r^2\right) - a^2 = \frac{\pi r^2}{2} - a^2$$

Step 2: Substitution [0.5 Mark]

$$\text{Area} = \frac{1}{2} \times \frac{22}{7} \times 49 - 49 = 77 - 49 = 28 \text{ cm}^2$$

Correct Option: (A) 28 cm^2

Area of shaded leaf = **28** cm^2

Q4. An automobile has two wipers of length 21 cm each, sweeping through an angle of 120°. If the blades do not overlap, the total area swept at each sweep is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 462 cm^2 (B) 924 cm^2 (C) 308 cm^2 (D) 616 cm^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Area Cleaned by One Wiper [0.5 Mark]

$$\text{Area}_1 = \frac{120^\circ}{360^\circ} \times \frac{22}{7} \times 21 \times 21 = \frac{1}{3} \times 22 \times 3 \times 21 = 462 \text{ cm}^2$$

Step 2: Total Area for Two Wipers [0.5 Mark]

$$\text{Total Area} = 2 \times 462 = 924 \text{ cm}^2$$

Correct Option: (B) 924 cm^2

Total Area Swept = **924** cm^2

Q5. In a circle of radius 6 cm, a chord subtends an angle of 60° at the centre. The area of the corresponding equilateral triangle formed by the radii and chord is: [1]

(A) $9\sqrt{3}$ cm^2 (B) $18\sqrt{3}$ cm^2 (C) $36\sqrt{3}$ cm^2 (D) $6\sqrt{3}$ cm^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Area Formula for Equilateral Triangle [0.5 Mark]

Since $OA = OB = 6$ cm and $\angle AOB = 60^\circ$, $\triangle OAB$ is equilateral with side $a = 6$ cm.

$$\text{Area} = \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4} \times (6)^2 = \frac{36\sqrt{3}}{4} = 9\sqrt{3} \text{ cm}^2$$

Correct Option: (A) $9\sqrt{3}$ cm^2

Area of equilateral triangle = **$9\sqrt{3}$** cm^2

Q6. **Assertion (A):** If the radius of a circle is doubled, its area becomes four times its original area.
Reason (R): The area of a circle varies directly as the square of its radius ($A = \pi r^2$). [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Verifying Assertion (A) [0.5 Mark]

Let original radius be r. If $r' = 2r$, then $A' = \pi(2r)^2 = 4\pi r^2 = 4A$. Thus (A) is **True**.

Step 2: Verifying Reason (R) [0.5 Mark]

Since $A = \pi r^2 \implies A \propto r^2$, doubling r multiplies area by $2^2 = 4$. (R) is **True** and explains (A).

Correct Option: **(A) Both (A) and (R) are true and (R) is the correct explanation of (A).**

Correct Option: (**A**)

SECTION B: Very Short Answer Type — MATHS MANTHAN | (2 × 2 = 4 Marks)

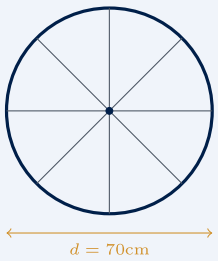
Q7. A bicycle wheel of diameter 70 cm rotates at 250 revolutions per minute. Find the speed of the cyclist in km/h. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Distance Covered in One Revolution [1 Mark]

Diameter $d = 70$ cm.

$$\text{Distance in 1 rev} = \pi d = \frac{22}{7} \times 70 = 220 \text{ cm} = 2.2 \text{ m}$$



Step 2: Speed Calculation in km/h [1 Mark]

$$\text{Distance in 1 hour} = 250 \times 2.2 \times 60 = 550 \times 60 = 33000 \text{ m} = 33 \text{ km}$$

$$\text{Speed} = 33 \text{ km/h}$$

Speed of cyclist = **33** km/h

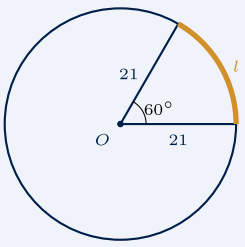
Q8. In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. Find the length of the arc. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: Radius $r = 21$ cm, $\theta = 60°$.

Step 1: Formula Declaration [1 Mark]

$$l = \frac{\theta}{360°} \times 2\pi r$$



Step 2: Substitution and Simplification [1 Mark]

$$l = \frac{60°}{360°} \times 2 \times \frac{22}{7} \times 21 = \frac{1}{6} \times 132 = 22 \text{ cm}$$

Length of the arc l = **22** cm

Q9. The length of the minute hand of a wall clock is 14 cm. Find the area swept by the minute hand in 20 minutes. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

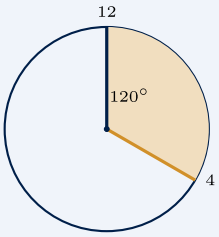
▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Radius $r = 14$ cm.

Step 1: Central Angle for 20 Minutes [1 Mark]

In 60 minutes, minute hand sweeps 360°:

$$\theta = \frac{360^\circ}{60} \times 20 = 120^\circ$$



Step 2: Area Calculation [1.5 Marks]

$$\text{Area} = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{120^\circ}{360^\circ} \times \frac{22}{7} \times 14 \times 14$$
$$= \frac{1}{3} \times 22 \times 2 \times 14 = \frac{616}{3} \approx 205.33 \text{ cm}^2$$

Step 3: Boxed Answer [0.5 Mark]

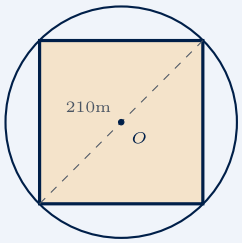
$$\boxed{\text{Area Swept} = \mathbf{205.33}\ \text{cm}^\mathbf{2} \quad \left(\text{or } \frac{\mathbf{616}}{\mathbf{3}}\ \text{cm}^\mathbf{2}\right)}$$

Q10. A circular field has a circumference of 660 m. A square sports plot of the largest possible area is inscribed in this field. Find the area of the square sports plot. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Radius of the Circular Field [1 Mark]

$$2\pi R = 660 \implies 2 \times \frac{22}{7} \times R = 660 \implies R = \frac{660 \times 7}{44} = 105 \text{ m}$$



Step 2: Diameter and Inscribed Square [1 Mark]

The largest square inscribed in a circle has its diagonal equal to the diameter:

$$\text{Diagonal } d = 2R = 2 \times 105 = 210 \text{ m}$$

Step 3: Area of the Square Sports Plot [1 Mark]

$$\text{Area} = \frac{1}{2}d^2 = \frac{1}{2} \times (210)^2 = \frac{44100}{2} = 22050 \text{ m}^2$$

$$\boxed{\text{Area of square plot} = \mathbf{22050}\ \text{m}^\mathbf{2}}$$

Q11. A chord of a circle of radius 15 cm subtends an angle of 60° at the centre.

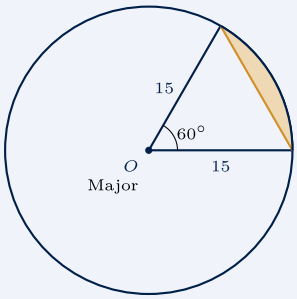
(a) Find the area of the corresponding minor segment of the circle. [2]

(b) Find the area of the corresponding major segment of the circle. [2]

(c) Determine the ratio of the area of the minor segment to that of the major segment. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$) [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Radius $r = 15$ cm, Central Angle $\theta = 60°$.



Part (a): Area of Minor Segment [2 Marks]

$$\text{Area of minor sector} = \frac{60°}{360°} \times 3.14 \times (15)^2 = \frac{706.5}{6} = 117.75 \text{ cm}^2$$

$$\text{Area}(\triangle OAB) = \frac{\sqrt{3}}{4} \times (15)^2 = \frac{1.73 \times 225}{4} = 97.3125 \text{ cm}^2$$

$$\text{Area of minor segment} = 117.75 - 97.3125 = 20.4375 \text{ cm}^2 \approx 20.44 \text{ cm}^2$$

Part (b): Area of Major Segment [2 Marks]

$$\text{Area of circle} = \pi r^2 = 3.14 \times (15)^2 = 706.5 \text{ cm}^2$$

$$\text{Area of major segment} = 706.5 - 20.4375 = 686.0625 \text{ cm}^2 \approx 686.06 \text{ cm}^2$$

Part (c): Ratio of Minor to Major Segment Areas [1 Mark]

$$\text{Ratio} = \frac{20.4375}{686.0625} = \frac{327}{10977} \approx 0.0298 \quad (\approx 1 : 33.57)$$

(a) Minor = **20.44** cm², (b) Major = **686.06** cm², (c) Ratio ≈ **0.0298**

Q12. Two circular flower beds are established on two opposite sides of a square lawn $ABCD$ of side 56 m. The centre of each circular flower bed is the intersection point O of the diagonals of the lawn.

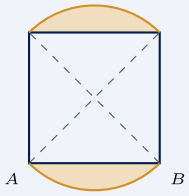
(a) Find the radius of each circular sector. **[1.5]**

(b) Calculate the total area of the entire garden (lawn + flower beds) using the sector method. **[2]**

(c) Verify the result by summing the area of the square and the two circular segment beds. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[1.5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Side of square lawn $a = 56$ m. Diagonals intersect at O with $\angle AOB = 90^\circ$.



Part (a): Radius of Circular Sector [1.5 Marks]

$$AC^2 = 56^2 + 56^2 = 2 \times 56^2 \implies AC = 56\sqrt{2} \text{ m}$$

$$r = OA = \frac{AC}{2} = 28\sqrt{2} \text{ m} \implies r^2 = 1568$$

Part (b): Total Area via Sector Method [2 Marks]

$$\text{Area of sector } OAB = \frac{90^\circ}{360^\circ} \times \frac{22}{7} \times 1568 = \frac{11}{14} \times 1568 = 1232 \text{ m}^2$$

$$\text{Area}(\triangle OBC) = \frac{1}{4} \times (56)^2 = 784 \text{ m}^2$$

$$\text{Total Area} = 2 \times 1232 + 2 \times 784 = 2464 + 1568 = 4032 \text{ m}^2$$

Part (c): Verification via Segments [1.5 Marks]

Area of 1 flower bed segment = Sector $OAB - \triangle OAB = 1232 - 784 = 448$ m^2

Total Area = Area of square $+ 2 \times$ (Segment bed) $= 3136 + 2(448) = 4032$ m^2

Both methods match identically.

(a) $r = \mathbf{28\sqrt{2}}$ m, (b) Total Area = **4032** m^2, (c) Verification: $3136 + 896 = \mathbf{4032}$ m^2

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Q13. Case Study: Antique Grandfather Clock Rotational Geometry

A horologist studies an antique grandfather clock. The minute hand has length 14 cm and the hour hand has length 7 cm. Both hands rotate smoothly about the common dial centre O.

(i) Calculate the angle through which the minute hand rotates in 25 minutes. [1]

(ii) Calculate the area swept by the minute hand between 8:00 am and 8:25 am. [1]

(iii) Calculate the area swept by the hour hand in 1 hour. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

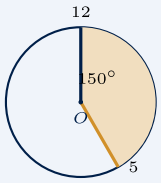
— OR —

Find the distance traversed by the tip of the minute hand in 45 minutes. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Given: Minute hand $r_m = 14$ cm, Hour hand $r_h = 7$ cm.

Part (i): Angle Rotated by Minute Hand in 25 Minutes [1 Mark]



$$\theta = \frac{360^\circ}{60} \times 25 = 6^\circ \times 25 = 150^\circ$$

Part (ii): Area Swept by Minute Hand (8:00 to 8:25 am) [1 Mark]

$$\text{Area} = \frac{150^\circ}{360^\circ} \times \frac{22}{7} \times 14 \times 14 = \frac{5}{12} \times 616 = \frac{770}{3} \approx 256.67 \text{ cm}^2$$

Part (iii): Area Swept by Hour Hand in 1 Hour [2 Marks]

In 1 hour, hour hand rotates through $\theta_h = \frac{360^\circ}{12} = 30^\circ$:

$$\text{Area} = \frac{30^\circ}{360^\circ} \times \frac{22}{7} \times 7 \times 7 = \frac{1}{12} \times 154 = \frac{77}{6} \approx 12.83 \text{ cm}^2$$

— OR (Alternative Choice) —

Distance Traversed by Tip in 45 Minutes: [2 Marks]

$$\theta = \frac{360^\circ}{60} \times 45 = 270^\circ$$

$$\text{Distance } l = \frac{270^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 14 = \frac{3}{4} \times 88 = 66 \text{ cm}$$

(i) $\theta = \mathbf{150^\circ}$, (ii) Area = **256.67** cm^2, (iii) Hour Area = **12.83** cm^2 (OR Tip Distance = **66** cm)

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